

AURA Forge

Governed Adaptive Engineering for FitGPT

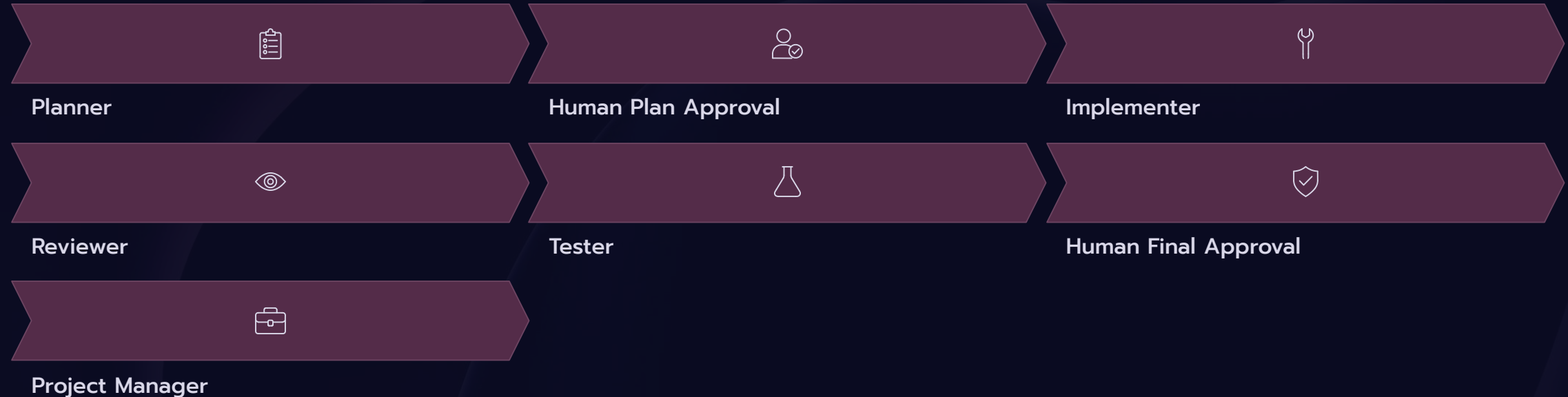
LaunchCode Agentic Engineering Capstone · **Muhammad Imran**

Give each software change only the autonomy it deserves.

THE PROBLEM

Fixed autonomy made simple changes too expensive

Every change — regardless of risk — received the same full workflow. The problem wasn't the pipeline. It was that **risk and governance were never matched**.



LOW Risk — Too Much Process

A documentation-only change still triggered every step.

- Unnecessary model calls
- Unnecessary human checkpoints
- Added cost and cycle time

HIGH Risk — Governance Is Necessary

Policy-sensitive work genuinely requires strong controls.

- Multiple human approvals justified
- Policy and evaluation gates required
- Full accountability trail needed

AURA Forge routes each change by risk

A deterministic classifier selects risk **before** any agentic route begins — no AI model, no tool grants, no code execution at this stage.



LOW gets speed. HIGH keeps governance.

AURA Forge removes overhead **only when the risk allows it**. More governance is correct when risk is higher — HIGH is not a failure, it is the right response.



LOW — Fastest Route

Implementer → Reviewer →
Deterministic Verification

0 human checkpoints



MEDIUM — Balanced Route

Implementer → Reviewer → Tester
→ Final Human Approval

1 human checkpoint



HIGH — Full Governance

Planner → Plan Approval →
Implementer → Reviewer → Tester
→ Policy/Evaluation Gates → Final
Human Approval → Project Manager

2 human checkpoints

The experiment compared matched scenarios

Control first. Freeze the router. Then measure AURA. Three locked scenarios on a 16-point quality rubric.

AF-LOW-001

Documentation / Accessibility
Representative LOW-risk work

AF-MEDIUM-001

Bounded Executable Test Change
Representative MEDIUM-risk work

AF-HIGH-001

Governance / MCP Policy-Sensitive Change
Representative HIGH-risk work

01

Unified Rubric

Apply locked 16-point quality rubric consistently across all scenarios.

03

Freeze AURA Router

Lock the AURA router configuration before taking any measurements to prevent drift.

02

Control Baseline

Measure the Control route first to establish a reliable baseline for comparison.

04

Separate Infrastructure Failures

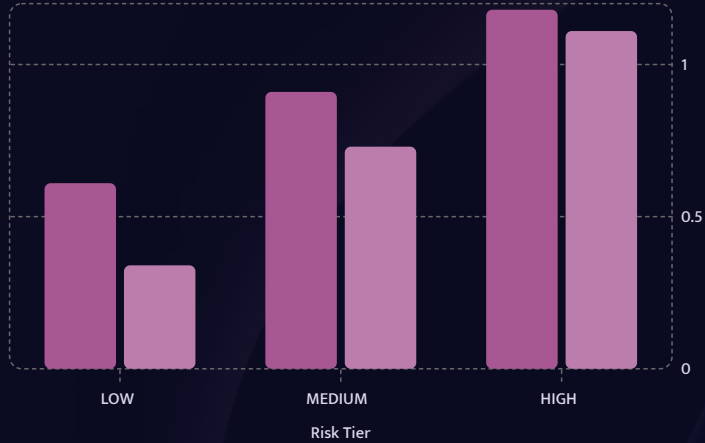
Keep infrastructure and provider failures isolated from product-quality scores.

MEASURED RESULTS

Less overhead. Stronger quality.

Measured across three representative capstone scenarios. All four aggregate metrics improved.

Pre-AURA Cost (\$) AURA Cost (\$)



19.22%

Cost Reduction

Successful-route cost across all tiers

50%

Fewer Checkpoints

Human approvals eliminated where safe

Cost by Risk Tier

LOW dropped 44%. MEDIUM dropped 20%. HIGH stayed close — full governance was intentionally retained.

33.33%

Fewer Model Roles

Where risk allowed reduction

48/48

Quality Score

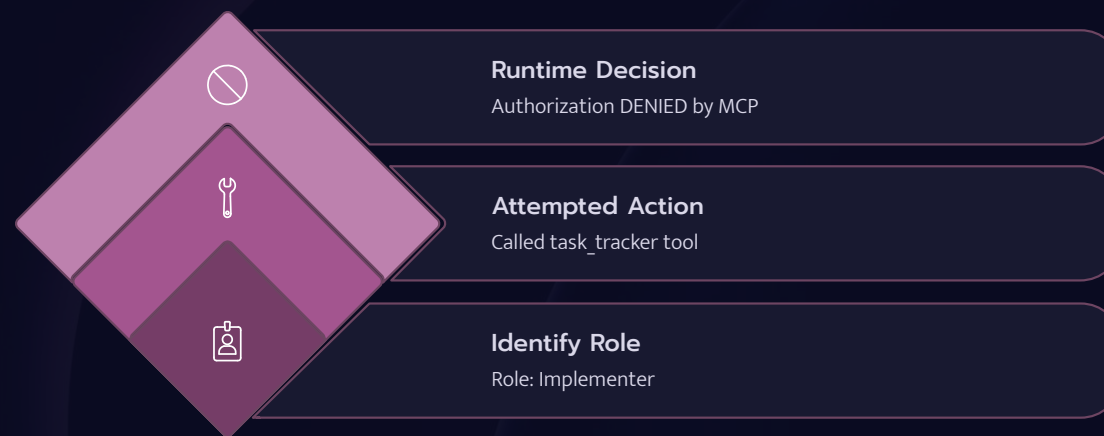
Up from 44/48 on the locked rubric

Source: measured AF-LOW / AF-MEDIUM / AF-HIGH control and AURA runs. Three representative scenarios only; not organization-wide.

Governance stopped a real overreach attempt

Policy was enforced by the runtime — not written in documentation. The Implementer attempted to use a tool reserved for the Project Manager.

Authorization Flow



Sanitized Evidence

Authorization error: role 'implementer' is not on the allow-list for task_tracker.

Allowed roles: ['project-manager']

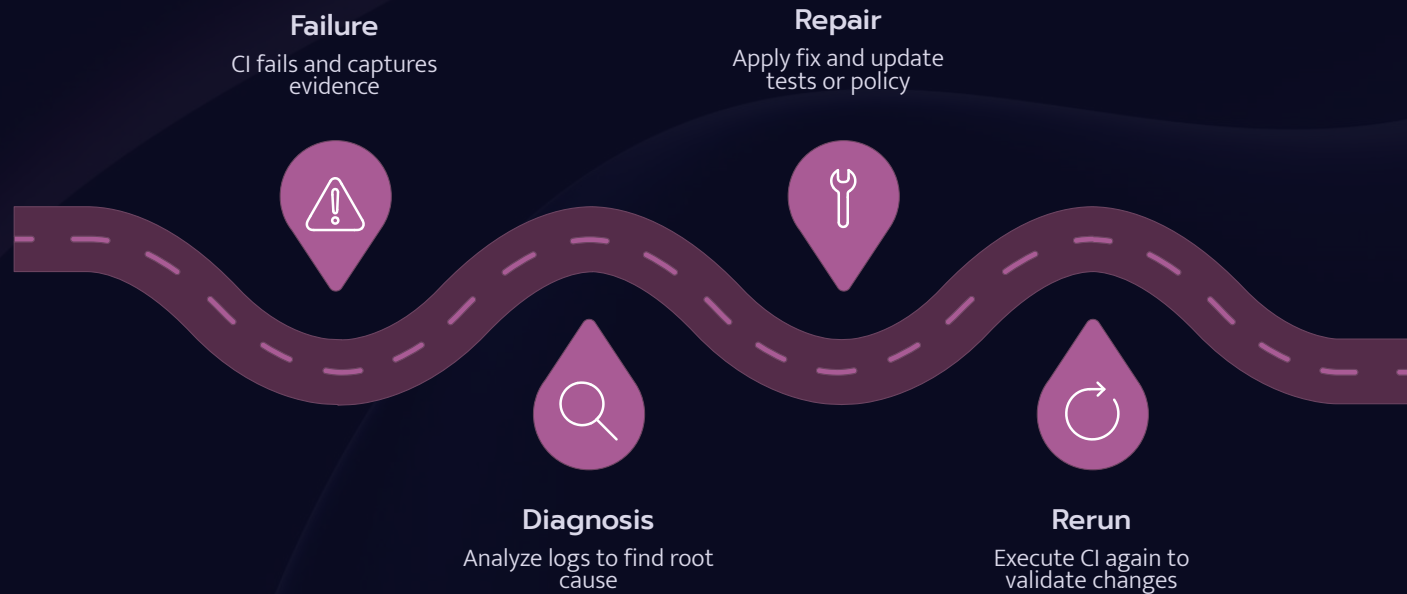
✓ External state changed: NO

✓ Model cost: \$0

The role that modifies work should not also declare it complete. Tool access is based on least privilege.

CI made the engineering gates visible

Failures were preserved, diagnosed, repaired, and rerun — evidence of a self-improving engineering loop, not a broken pipeline.



Permanent Gates

 Change Classifier	 Policy Tests
 Evaluation Gate	 Pipeline Integrity
 Audit Trail	 Advisory Review

Verified Final CI Evidence

GitHub Actions Run 31531270032 — SUCCESS

Change Classifier — PASS	Policy Tests — PASS
Evaluation Gate — PASS	Pipeline Integrity — PASS
Audit Trail — PASS	Advisory AI — safely skipped

A controlled contract fault was introduced, detected, preserved, and repaired.

Change Passport proves readiness

Readiness is backed by machine-generated evidence — not a claim. **If evidence does not exist, the Passport does not invent it.**

Change Passport Fields

Risk Tier	LOW / MEDIUM / HIGH
Selected Route	Adaptive path assigned
Roles Executed	Scoped per tier
Test Result	Pass / Fail
Policy Result	Pass / Fail
Evaluation Result	Score on rubric
Human Approvals	Recorded per gate
CI Status	Green / Failed
Readiness	Confirmed / Blocked

Right-Tool Decision

Agentic auditor → stable, factual, predictable task → replaced with deterministic code.

📄 **\$0 model cost · ≈ 0.156 ms** local runtime

Not every engineering task deserves an AI agent.

Impact is measured, bounded, and honest

Measured Impact

- Quality improved: 44/48 → 48/48
- LOW and MEDIUM overhead dropped
- HIGH governance stayed in place
- Successful-route cost fell across all three scenarios
- Fewer model roles and human checkpoints where risk allowed

Limitations

- Three representative scenarios only
- No production deployment or real-user data
- Advisory AI unavailable in CI — safely skipped
- Results are not organization-wide projections

The best agentic engineering system is not the one that uses the most agents. It is the one that knows when not to use them.

AURA Forge gives each change the minimum autonomy it deserves, then proves the route stayed inside policy.